Human Resources Mangament System

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1. Introduction

1.1 Purpose

The purpose of this document is to specify the requirements for the Human Resource Management System (HRMS). This system is designed to automate and streamline various HR functions such as employee management, payroll, attendance, recruitment, performance appraisal, and leave management.

1.2 Scope

The HRMS will support the following key functionalities:

- Employee information management
- Attendance and leave tracking
- Payroll processing
- Recruitment and on-boarding
- Performance appraisal
- Role-based access and reporting

2. Overall Description

2.1 Product Perspective

This system is a standalone web-based application. It will interface with a backend database and provide a user-friendly GUI for HR staff, employees, and administrators.

2.2 Product Functions

- Maintain employee profiles
- Track attendance and leaves
- Generate payslips and tax calculations
- Post and manage job vacancies
- Evaluate employee performance

2.3 User Classes and Characteristics

- **Admin**: Full access to the system, including all modules and reports.
- **HR Staff**: Manage employee data, payroll, and recruitment.
- **Employees**: View their profiles, request leaves, view payslips.
- **Managers**: Approve leaves, conduct performance reviews.

3.Function Requiremnets

3.1. Employee Management

- Stores and manages employee details like name, ID, department, contact info, joining date, etc. (Add/edit/delete employee information)
- Tracks employment history, documents, and job roles(Assign departments, roles, and managers)
- Store documents (CV, certificates)

3.2. Attendance Management

- Records daily attendance (check-in/check-out times).
- Tracks absences, late arrivals, and working hours.
- Can integrate with biometric or RFID systems.
- Generate attendance reports

3.3. Leave Management

- Employees can request different types of leave (e.g., vacation, sick leave).
- Managers approve or reject leave requests.
- Tracks leave balances and leave history.
- Holiday calendar integration

3.4. Payroll Management

- Calculates employee salaries based on attendance, leave, bonuses, and deductions.
- Generates payslips.
- Handles tax, insurance, and other statutory deductions.

3.5. Recruitment & On-boarding

- Posts job openings.
- Tracks applicants and interview stages.(schedule)
- Manages on-boarding tasks for new hires (document submission, orientation, etc.).

3.6. Performance Management

- Sets goals or KPIs for employees.
- Allows managers to review and rate performance.(Conduct reviews)
- Helps in promotions, bonuses, and appraisals for reports

3.7. Reports & Analysis

- Generates reports on attendance, payroll, headcount, and custom.
- Helps HR make data-driven decisions.
- [Employee headcount,Departmental summaries ,Payroll summaries]

3.8. Role-Based Access & Security

- Provides different access levels (Admin, HR, Employee, Manager).
- Ensures data privacy and secure login.

3.9.Notifications and Alerts

- Email and in-system notifications for approvals, deadlines

4. Non-Functional Requirements

4.1 Performance Requirements

- Support for 500+ concurrent users and Response time < 2 seconds for most operations

4.2 Security Requirements

- Role-based access control and Encrypted passwords and sensitive data and Audit logs

4.3 Usability

- Intuitive and user-friendly interface

4.4 Reliability and Availability

- 99.9% uptime and Automatic backups

4.5 Maintainability

- Modular codebase with Clear documentation

4.6 Portability

- Deployable on Linux or Windows servers and Cloud-ready

5. System Features

- Multi-role login
- Search and filter functionalities
- Export data (Excel, PDF)
- Dashboard with KPIs

6. Use Case Modeling

6.1 Use Case: Employee Login

Actor: Employee

Precondition: Valid login credentials

Description: Employee logs into the system to view personal data, payslips, or apply for leave

6.2 Use Case: HR Approves Leave

Actor: HR Manager

Precondition: Employee has applied for leave

Description: HR views leave request and approves/rejects it

7. Data Requirements

7.1 Data Dictionary

Employee Table: EmpID, Name, Email, Department, etc.

Leave Table: LeaveID, EmpID, StartDate, EndDate, Status

Payroll Table: PayrollID, EmpID, Month, Amount, Deductions

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8.Technologies Used – MERN Stack

8.1 Front-End Technologies

- **React.js/Next.js** – For building a dynamic, component-based single-page application (SPA)
- **Bootstrap / Tailwind CSS** – For responsive and modern UI design.
- **React Router** – For client-side routing/navigation.

8.2 Back-End Technologies- (Prisma) or Node.js framework

- **Node.js** – JavaScript runtime for server-side scripting.
- **Express.js** – Web framework for building RESTful APIs.
- **REST API** – For communication between front-end and back-end.
- **Multer** – For handling file uploads (e.g., employee documents).

8.3 Database

- **MongoDB** – NoSQL document-based database for storing employee records, leaves, payroll data, etc.
- **MYSQL/SQLITE**

8.4 Authentication & Security

- **JWT (JSON Web Tokens)** – For user authentication and secure session handling.
- **BCrypt.js** – For hashing passwords before storing them in the database.
- **Helmet.js** – For securing HTTP headers.
- **CORS** – For handling cross-origin requests between client and server.
- **HTTPS / SSL** – For secure data transmission.

8.5 DevOps & Deployment

- **CI/CD** – GitHub Actions or GitLab CI for continuous integration and deployment.
- **Cloud Hosting**:

  Frontend: Vercel / Netlify / Firebase Hosting

  Backend: Render / Railway / Heroku / AWS EC2

  Database: MongoDB Atlas (cloud-hosted MongoDB)

8.6 Version Control

- **Git** – For tracking code changes.
- **GitHub / GitLab** – For code repository, collaboration, and issue tracking.

8.7 Testing Tools

- **Postman** – For testing API endpoints.
- **Jest** – For unit testing.
- **Supertest** – For testing Node.js/Express routes.
- **React Testing Library** – For front-end component testing.

8.8 Optional Tools & Libraries

- **Chart.js / Recharts** – For dashboard data visualization (e.g., leave trends, performance scores).
- **Cloudinary / Firebase Storage** – For storing and retrieving uploaded documents/photos.
- **Socket.io** *(optional)* – For real-time updates (e.g., notifications).
- **Email API/Email.js** – Nodemailer or integration with third-party services like SendGrid for sending automated emails.